

Chapter 146 — Pathophysiology, Evaluation and Medical Management of Adrenal Disorders — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

1. **Test every mass >1 cm:** cortisol + metanephrines; aldosterone if hypertensive/hypokalaemic.
2. **Unenhanced computed tomography (CT) <10 Hounsfield units (HU)** → lipid-rich adenoma.
3. **15-min washout:** absolute >60% or relative >40% → adenoma.
4. **Resect if >4 cm** (not myelolipoma), suspicious imaging, or growth >1 cm.
5. **Exclude pheochromocytoma before biopsy, sampling or surgery** — surge can kill.
6. **Primary aldosteronism (PA):** aldosterone-to-renin ratio (ARR) → confirm → adrenal vein sampling (AVS).
7. **1-mg dexamethasone test:** cortisol ≤1.8 µg/dL excludes; >5.0 = autonomous.
8. **Pheochromocytoma prep:** α-blockade 7–14 days; β only after α.
9. **Adrenocortical carcinoma (ACC):** open if >6 cm/invasive; no preoperative biopsy.
10. **Prior contralateral nephrectomy** → check remaining adrenal; crisis risk.

EMBRYOLOGY AND ANATOMY

- **Veins:** right short → inferior vena cava; left long → left renal (+ inferior phrenic).
- **Anomalous venous drainage:** up to 9% — matters for AVS.
- **Trap:** testicular adrenal rests in congenital adrenal hyperplasia mimic testicular tumour.

ADRENAL PHYSIOLOGY

- **Zones (out → in):** glomerulosa aldosterone; fasciculata cortisol; reticularis androgens.
- **Aldosterone driven by angiotensin II + K⁺** → glomerulosa survives pituitary failure.
- **Medulla:** epinephrine 80%, norepinephrine 19%, dopamine 1%.
- **Phenylethanolamine-N-methyltransferase (PNMT)** (norepinephrine → epinephrine) essentially adrenal-only; glucocorticoid-potentiated.
- **>90% of circulating metanephrine is adrenal** → basis of diagnosis.

EVALUATION OF ADRENAL LESIONS

- **Gold standard first test:** unenhanced CT; ≤0 HU 100% specific.
- **Gold standard characterisation:** CT washout (unenhanced 5 mm → 60 s → 15 min).
- **Chemical-shift magnetic resonance imaging (MRI) drop-out** ≈ unenhanced CT; gadolinium washout not used.
- **Pitfalls:** up to 7% of metastases <10 HU; renal cell (RCC) and hepatocellular (HCC) carcinoma metastases wash out.
- **Malignancy by size:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm ≈20%.
- **<10 HU:** no malignancy regardless of size (largest series).
- **Macroscopic fat (<-30 HU)** = myelolipoma; >50% fat typical.
- **Follow-up (European Society of Endocrinology-European Network for the Study of Adrenal Tumors, ESE-ENSAT):** <4 cm, benign, silent → none; others 6–12 months.
- **Growth concern:** >20% diameter and ≥5 mm (classic: 1 cm).
- **Pheochromocytoma staging:** fluorodeoxyglucose positron emission tomography (FDG PET) gold standard; multiple endocrine neoplasia type 2 (MEN-2) → metaiodobenzylguanidine (MIBG).
- **MIBG cold** in von Hippel-Lindau (VHL) and succinate dehydrogenase B (SDHB); FDG PET sensitive there.
- **Incidentaloma function:** ≈11% active; ≈18% potentially surgical.
- **Screen for PA if:** ≥3 drugs incl. diuretic; ≥4 drugs; ≥150/100 ×3.

- **Hypertension + hypokalaemia, incidentaloma + hypertension, onset <20, stroke <50.**
- **ARR:** 8–10 am seated; renin floor 0.2 ng/mL/h.
- **Cut-offs:** ARR >20–30; plasma aldosterone >15–20 ng/dL.
- **Stop:** spironolactone/amiloride 4–6 weeks; angiotensin-converting enzyme (ACE) inhibitors/angiotensin receptor blockers (ARBs) may continue.
- **β-blocker** → false-positive ARR; ACE inhibitor/ARB → false negative.
- **Undetectable renin on ACE inhibitor/ARB** → confirm despite normal ARR.
- **Skip confirmation:** spontaneous hypokalaemia + undetectable renin + aldosterone >20.
- **IV saline 2 L/4 h:** aldosterone >5 = PA; >10 suggests adenoma.
- **Oral salt:** urine Na >200 mmol; aldosterone >12 µg/day = PA.
- **AVS:** selectivity 1.1–5:1; lateralisation 2–4:1 (higher with cosyntropin).
- **Skip AVS:** age <40 + unilateral adenoma >1 cm + normal contralateral.
- **Cortisol first line:** 1 mg dexamethasone 11 pm → cortisol 8–9 am.
- **Pregnancy** → urinary free cortisol; epilepsy → salivary; renal failure → dexamethasone test.
- **Oral contraceptive** → up to 50% false-positive dexamethasone test.
- **Late-night salivary cortisol:** ≤145 ng/dL (mass spectrometry).
- **Commonest cause of hypercortisolism:** exogenous steroid.
- **Gold standard pituitary vs ectopic:** inferior petrosal sinus sampling + corticotropin-releasing hormone (CRH).
- **First line pheochromocytoma:** plasma free (supine 20 min) or urinary fractionated metanephrines.
- **Plasma upper limits:** normetanephrine 0.61, metanephrine 0.31 nmol/L.
- **Diagnostic:** normetanephrine >2.2 or metanephrine >1.2 nmol/L.
- **Stop:** paracetamol ≥5 days; tricyclics; phenoxybenzamine.
- **Plasma free metanephrines:** sensitivity 97–99% (Lenders multicentre study).
- **Biopsy only if:** known cancer + result changes management.
- **Never biopsy:** suspected ACC (seeding); unexcluded pheochromocytoma.

ADRENAL NEOPLASIA

- **Incidentaloma:** >1 cm; ≈5% of scans; up to 7% at ≥70.
- **Commonest:** adenoma (>85%); ≈93% of adenomas inactive.
- **PA commonest secondary hypertension:** 3–13% primary care; 30% tertiary.

- PA subtypes: bilateral hyperplasia 60%; adenoma 35%.
- PA vs essential hypertension: atrial fibrillation $\times 12$, myocardial infarction $\times 6.5$, stroke $\times 4$.
- Familial hyperaldosteronism (FH) I: CYP11B1/CYP11B2 chimera $\rightarrow$ glucocorticoid-remediable; FH-III: KCNJ5.
- After adrenalectomy for PA: biochemical cure $\approx 94\%$; drug-free 37–52%.
- Medical PA: spironolactone 25–400 mg; eplerenone 25–100 mg (selective).
- Post-op insufficiency: mild autonomous cortisol $\approx 65\%$; overt Cushing $\approx 99.7\%$.
- Cover: hydrocortisone 100 mg IV day of surgery, then taper.
- Adrenal recovery: 6.5 months (mild) vs 11 months (Cushing).
- Pheochromocytoma: $\approx 5\%$ of incidentalomas; $\approx 0.5\%$ of hypertension.
- Hereditary $\approx 1\%$; 24% of “sporadic” carry mutation $\rightarrow$ offer testing to all.
- Highest malignancy: SDHB (30–50%); MEN-2 almost always adrenal.
- VHL: normetanephrine only (metanephrine normal).
- Malignant = metastases only; extra-adrenal $> 1/2$ malignant.
- Triad: headache, sweating, tachycardia; paroxysmal hypertension only 30–50%.
- CT: > 10 HU (mean ≈ 35), no washout; iodinated contrast safe.
- Phenoxybenzamine: 10 mg twice daily, 7–14 days, ≈ 1 mg/kg.
- Target: 120–130/80 mmHg seated; mild postural drop acceptable.
- Metyrosine blocks tyrosine hydroxylase: 1.5–2 g/day.
- Post-op: hypotension; rebound hypoglycaemia.
- Follow-up: metanephrines at 6 months, then annually lifelong; recurrence up to 16%.
- Recurrence risk: extra-adrenal hazard ratio (HR) 11.2, hereditary 3.4, right-sided 3.1.
- MEN-2/VHL $\rightarrow$ cortical-sparing adrenalectomy.
- Metastatic: ^{131}I -MIBG (US Food and Drug Administration approval 2018); cyclophosphamide–vincristine–dacarbazine (CVD); 5-year survival $\approx 50\%$.

NONFUNCTIONAL BENIGN LESIONS

- Myelolipoma: operate only if large/symptomatic; proposed 6–10 cm cut-offs.
- Smallest bleeding myelolipoma 6.3 cm; bleeders average 10–14 cm.
- Cyst: pseudocyst carries malignancy ($\approx 7\%$ of cysts).
- Excise cyst if: > 5 cm, thick wall, heterogeneous, symptomatic.
- Aspirate cyst only after excluding pheochromocytoma.
- Ganglioneuroma: young; encases vessels without narrowing; $\leq$ muscle density.
- Adrenal oncocytoma: $> 30\%$ malignant — unlike kidney.

MALIGNANT ADRENAL NEOPLASMS

- ACC: 0.7–2 per million; bimodal (first decade, 40s–50s).
- $> 90\%$ exceed 5 cm; mean 10–12 cm; unenhanced ≈ 39 HU.
- Functional: 50–79% adults, $> 90\%$ children; cortisol commonest.
- Feminisation in a man $\rightarrow$ carcinoma.
- Weiss: ≥ 3 of 9 criteria = malignant; > 5 mitoses/50 high-power fields.
- American Joint Committee on Cancer (AJCC): I ≤ 5 cm, II > 5 cm confined; ENSAT IV = distant only.
- Stage IV ≈ 40 –50% at diagnosis; 5-year survival $< 5\%$.
- Recurrence after resection: 60–80%.

- Cytoreducer if: $> 90\%$ of disease removable; else systemic therapy first.
- FIRM-ACT: mitotane + etoposide–doxorubicin–cisplatin (M-EDP) first line — better response/progression-free survival, not overall survival.
- ADJUVO: no benefit of adjuvant mitotane (R0 stage I–III).
- Adjuvant radiotherapy: $\downarrow$ local recurrence, no survival gain.
- Surveillance: CT 3-monthly 2–3 years, then 6-monthly to 5.
- Paediatric ACC: virilisation; Li–Fraumeni (TP53); stage split 200 g.
- Known cancer: $> 50\%$ of new adrenal lesions are metastases.
- Adrenalectomy at nephrectomy only if: abnormal adrenal or thrombus to adrenal vein.

NON-NEOPLASTIC ADRENAL DISORDERS

- Cushing causes: pituitary $\approx 70\%$; ectopic 10–15%; adrenal $\approx 10\%$.
- Trans-sphenoidal cure: 60–80%; macroadenoma $< 15\%$.
- Bilateral adrenalectomy: after failed pituitary treatment or life-threatening disease.
- Nelson syndrome: 8–29% after bilateral adrenalectomy — pituitary growth.
- Ectopic adrenocorticotrophic hormone (ACTH) commonest: bronchial carcinoid; pheochromocytoma can cause.
- ACTH-independent macronodular hyperplasia: huge nodular glands (McCune–Albright).
- Primary pigmented nodular disease: small pigmented glands (Carney complex).
- Cushing urology: urolithiasis up to 50%; hypogonadotropic hypogonadism.
- Addison commonest: autoimmune (West); tuberculosis (developing).
- Secondary insufficiency: aldosterone preserved $\rightarrow$ no fludrocortisone.
- Primary only: hyperpigmentation, hyperkalaemia, salt craving.
- Stress dose: minor 25; moderate 50–75; major 50 mg 8-hourly 48–72 h.
- Hydrocortisone 20 $\approx$ prednisone 5 $\approx$ dexamethasone 0.5 mg.
- Congenital adrenal hyperplasia: 21-hydroxylase $> 95\%$; don't screen incidentalomas.

CLASSIC TRAPS

- Trap: PA needs hypokalaemia — **Actually:** most are normokalaemic.
- Trap: calcification means benign — **Actually:** up to 30% of ACCs calcify.
- Trap: β -blocker first for pheochromocytoma tachycardia — **Actually:** never before α -blockade.
- Trap: IV contrast triggers crisis — **Actually:** no evidence; safe.
- Trap: CT nodule decides PA side — **Actually:** AVS; $\approx 20\%$ of adenomas < 1 cm.
- Trap: histology proves malignant pheochromocytoma — **Actually:** only metastases.
- Trap: MIBG best for all — **Actually:** FDG PET, except MEN-2.
- Trap: stop ACE inhibitors before ARR — **Actually:** stop spironolactone/amiloride instead.
- Trap: adrenal metastases cause Addison — **Actually:** need bilateral, bulky > 4 cm.
- Trap: < 10 HU always benign in known cancer — **Actually:** up to 7% of metastases are < 10 HU.